Problem Set 1 — Equilibrium bond length of H_2 with DFT

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October 26, 2026

Method

Total energies of H_2 were computed with `pw.x` (Quantum ESPRESSO [1]) using the supplied input `H2.in`. The numerical settings were kept fixed: 12 Å cubic box, 60 Ry plane-wave cutoff, Γ point only. Only the bond length b (z-coordinate of the second H atom) was varied, from 0.60 to 1.20 Å in steps of 0.04 Å, and for part (c) the pseudopotential was changed from `H.pz-vbc.UPF` (LDA) to `H.pbe-vbc.UPF` (GGA–PBE [2]). All SCF calculations converged.

(a) Energy versus bond length

The LDA total energies are listed in Table 1 (PBE values from part (c) alongside) and plotted in Fig. 1(a). The lowest LDA total energy on the grid is at

$$b = 0.76\,\text{Å}, \qquad E = -2.272\,005\,15\,\text{Ry}.$$

(LDA and PBE total energies are each defined only within their own functional and pseudopotential and are not compared with each other.)

Table 1: Total energy of H_2 versus bond length from `pw.x` (60 Ry, 12 Å box, Γ point). Lowest grid point in bold.

b (Å)	E_{LDA} (Ry)	E_{PBE} (Ry)	b (Å)	E_{LDA} (Ry)	E_{PBE} (Ry)
0.60	−2.223 521 48	−2.283 701 02	0.92	−2.251 200 42	−2.300 677 64
0.64	−2.246 711 46	−2.305 034 90	0.96	−2.241 282 85	−2.289 884 21
0.68	−2.261 317 88	−2.318 011 91	1.00	−2.230 300 85	−2.278 109 58
0.72	−2.269 249 15	−2.324 479 71	1.04	−2.218 515 10	−2.265 607 94
0.76	**−2.272 005 15**	**−2.325 893 17**	1.08	−2.206 138 39	−2.252 584 17
0.80	−2.270 758 72	−2.323 404 42	1.12	−2.193 349 16	−2.239 209 53
0.84	−2.266 426 64	−2.317 924 29	1.16	−2.180 300 16	−2.225 630 23
0.88	−2.259 723 01	−2.310 165 18	1.20	−2.167 122 26	−2.211 972 18

(b) Quadratic fit

Near the minimum the curve was fitted with $E(b) = \alpha + \beta b + \gamma b^2$; setting $\mathrm{d}E/\mathrm{d}b = \beta + 2\gamma b = 0$ gives $b_{\text{eq}} = -\beta/(2\gamma)$. The fit used `numpy.polyfit(b,E,2)`, which returns the coefficients highest power first, $[\gamma, \beta, \alpha]$. A parabola is only a valid model close to the minimum (the curve is steep on the compressed side and flattens towards dissociation), so I used the five points within ±0.10 Å of the lowest grid point, $b = 0.68$–0.84 Å: the smallest symmetric window that still over-determines the three parameters. Table 2 shows that wider windows pull b_{eq} systematically outward; the spread between the 3-, 5- and 7-point fits, ±0.005 Å, is taken as the uncertainty of b_{eq}. For LDA:

$$\alpha = -1.514\,102\,\text{Ry}, \quad \beta = -1.965\,223\,\text{Ry/Å}, \quad \gamma = 1.273\,622\,\text{Ry/Å}^2, \qquad b_{\text{eq}}^{\text{LDA}} = -\frac{\beta}{2\gamma} = 0.772\,\text{Å}.$$

(c) GGA–PBE and comparison with LDA and experiment

With `H.pbe-vbc.UPF` the lowest grid point is again $b = 0.76$ Å ($E = -2.325\,893\,17$ Ry, Table 1, Fig. 1). The same five-point fit gives

$$\alpha = -1.611\,497\,\text{Ry}, \quad \beta = -1.883\,171\,\text{Ry/Å}, \quad \gamma = 1.240\,985\,\text{Ry/Å}^2, \qquad b_{\text{eq}}^{\text{PBE}} = 0.759\,\text{Å}.$$

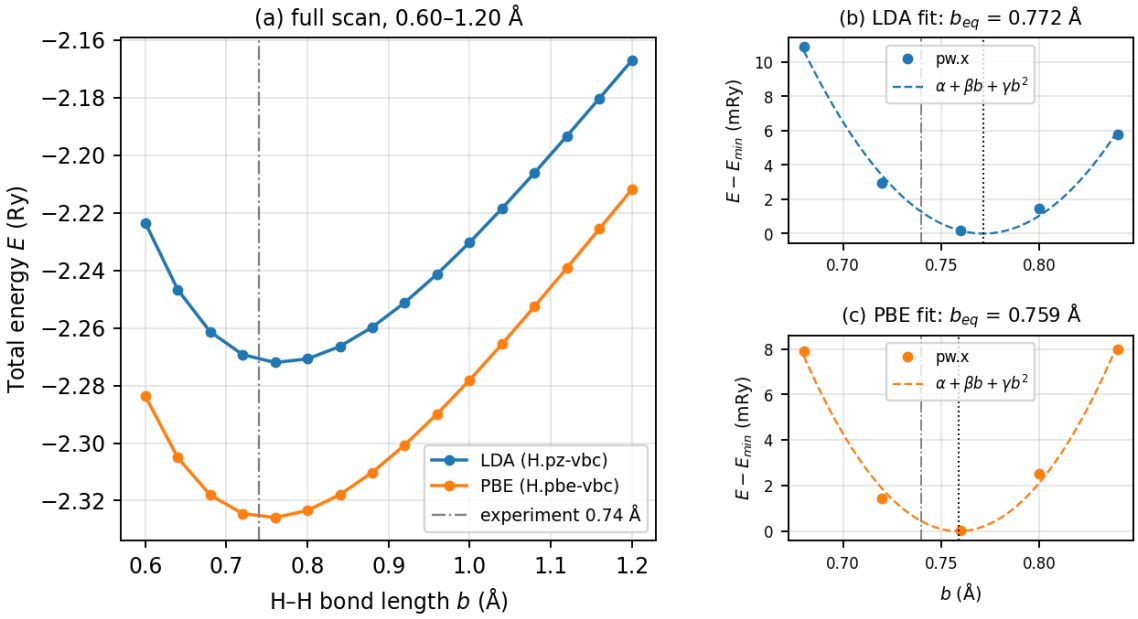


Figure 1: (a) Total energy of H_2 versus bond length for LDA (`H.pz-vbc`) and PBE (`H.pbe-vbc`); the dash-dotted line marks the experimental bond length. (b), (c) The five points used in the quadratic fit for LDA and PBE, relative to the fitted minimum, with the fitted parabola (dashed) and b_{eq} (dotted).

Table 2: Sensitivity of the fitted b_{eq} to the fit window (half-width around $b = 0.76$ Å). The window used is in bold.

half-width (Å)	points	$b_{\mathrm{eq}}^{\mathrm{LDA}}$ (Å)	$b_{\mathrm{eq}}^{\mathrm{PBE}}$ (Å)
0.06	3	0.7675	0.7545
0.10	**5**	0.7715	0.7587
0.14	7	0.7773	0.7650
0.30	12	0.7990	0.7830

A check with a 100 Ry cutoff and a 16 Å box changed both b_{eq} values by less than 0.001 Å.

Table 3: Equilibrium bond length of H_2 compared with the experimental value 0.74 Å.

Method	b_{eq} (Å)	error vs. 0.74 Å
LDA (`H.pz-vbc.UPF`)	0.772 ± 0.005	+4.3 %
PBE (`H.pbe-vbc.UPF`)	0.759 ± 0.005	+2.5 %
Experiment	0.74	—

Both functionals overestimate the bond length; PBE reduces the error from 4.3 % to 2.5 %. LDA evaluates the exchange–correlation energy at each point as if the density there were a uniform electron gas, which is a poor description of H_2, whose density is strongly peaked at the two nuclei and decays exponentially away from them. PBE adds a dependence on the density gradient $|\nabla n|$, which describes such inhomogeneous densities better and brings the bond length closer to experiment. The PBE curvature γ is also slightly smaller, i.e. the bond is marginally softer. The remaining error reflects the approximate functional and pseudopotential.

References

[1] P. Giannozzi *et al.*, J. Phys.: Condens. Matter **21**, 395502 (2009); **29**, 465901 (2017).

[2] J. P. Perdew, K. Burke and M. Ernzerhof, Phys. Rev. Lett. **77**, 3865 (1996).